



Final Exam in
TDT4125 ALGORITHM CONSTRUCTION, ADVANCED COURSE

Saturday May 24, 2014
Time: 09:00–13:00

Aids: (B) All printed/handwritten; specific, simple calculator

Language: English

Please read the entire exam before you start, plan your time, and prepare any questions for when the teacher comes to the exam room. Make assumptions where necessary. Keep your answers short and concise. Long explanations that do not directly answer the questions are given no weight.

All subtasks count equally toward the final score.

Problem 1

- a) Briefly discuss the idea of a relaxation, as it applies to the set of feasible solutions. How could the same concept be applied to what constitutes a valid instance of a problem? What would the difference be? How does this latter form of relaxation relate to reductions?
- b) How is the idea of relaxation used when constructing approximation algorithms?
- c) When dealing with linear programming, a straightforward approach to relaxation can sometimes lead to a loss of too much structure or information. Discuss briefly a common way to mitigate this, and to retain some information about what has been removed, but in a less strict manner.
- d) How can one safely remove the randomness from a randomized approximation algorithm?
- e) Assume that you want to minimize the objective $\max_{i=1 \dots n} c_i x_i$, where the coefficients c_i are all different, and are all positive integers, and you have the constraint $x_i \in \{0, 1\}$, for $i = 1 \dots n$. Show how you could reduce this to the objective of minimizing $\sum_{i=1}^n c'_i x_i$, with a different set of coefficients c' , so that the resulting solution for x will be optimal for your original problem, regardless of other constraints.

Consider the following integer programming formulation of the traveling salesman problem. The graph $G = (V, E)$ has nodes corresponding to the cities, and edges corresponding to the distances between them. Let $\delta(S)$ be the subset of edges in E that have exactly one endpoint in S . The variable x_e in the integer formulation is 1 if the edge is part of the tour, and 0 otherwise.

$$\begin{aligned}
 \min \quad & \sum_{e \in E} c_e x_e \\
 \text{s.t.} \quad & \sum_{e \in \delta(S)} x_e \geq 2, \quad \forall S \subset V, S \neq \emptyset, \\
 & x_e \in \{0, 1\}, \quad \forall e \in E.
 \end{aligned}$$

- f) Though you have no guarantees for the quality of the results, you would like to run some experiments using a linear programming relaxation of the program. Explain how you would do this.

The *min-max TSP problem* is a version of the TSP problem where we do not wish to minimize the *sum* of the distances in the tour, but rather we wish to minimize the magnitude of the greatest of those distances. In other words, we wish to minimize

$$\max\{c_{k(n)k(1)}, c_{k(1)k(2)}, \dots, c_{k(n-1)k(n)}\},$$

where c_{ij} is the distance between cities i and j and $k(i)$ is the i th city in the tour.

- g) Rewrite the integer program for TSP so that it is an integer program for the min-max TSP problem. Note that the method from d) will not work here.
- h) Show that min-max TSP is NP-hard, even if our cost function c is metric, that is, $c_{ik} \leq c_{ij} + c_{jk}$.
- i) What approximation bound would the double-tree algorithm give for the min-max TSP problem for metric inputs? Explain your reasoning clearly and thoroughly.

Now consider the case of so-called *ultrametric* distances. This means that our costs now obey the so-called *strong* triangle inequality:

$$c_{ik} \leq \max\{c_{ij}, c_{jk}\}$$

Assume that you modify Christofides' algorithm for the min-max problem in the following way: You use some given black-box procedure that can efficiently find both a min-max spanning tree (a spanning tree whose maximum edge is minimum) and a min-max matching (a matching whose maximum edge is minimum). You now use these instead of the MST and min-cost matching subroutines in Christofides' algorithm.

- j) What approximation bound would this modification of Christofides' algorithm give for the min-max TSP problem for ultrametric inputs? Explain your reasoning clearly and thoroughly.